

Project Design Phase Proposed Solution Template

Date	01 October 2026
Team ID	SWTID-2026-2051
Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Solution Template:

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	In ServiceNow, a user who has access to a table can normally see every record in it. In an institution, this means student details such as email, phone number and faculty information from all branches (ECE, EEE, CSE) are exposed to users who only need their own branch's data. There is also no separation between who can view, create, edit and delete records. Institutions need record-level security that depends on the field value (Branch) and on the user's role.
2	Idea / Solution description	Implement script-controlled Access Control Lists (ACLs) on the custom table <code>u_institution_details</code> . Custom roles control each operation: <code>bb1</code> for Read, <code>bb2</code> for Create, <code>bb3</code> for Write and <code>bb4</code> for Delete. The Read ACL combines a data condition (Branch is EEE) with an advanced script: admins get full access, users with <code>bb1</code> see only EEE records, and everyone else is denied. The solution is verified by impersonating the EEE user, a user without the roles, and an admin.
3	Novelty / Uniqueness	Combines a field-based data condition with an advanced script in a single ACL, so access is decided by both the user's role and the record's content. Each operation has its own role, which follows the least-privilege principle. Admin override is kept, no table is duplicated per branch, and no external tools or plugins are needed: it runs fully on native ServiceNow features.
4	Social Impact / Customer Satisfaction	Protects students' personal data and privacy, and builds trust between students, faculty and the institution. It lowers the risk of accidental or intentional misuse of data and supports data-privacy and audit requirements. Branch staff get a clean view showing only the records relevant to them, and admins keep full control, which improves satisfaction for both.
5	Business Model (Revenue Model)	This is an internal security feature on the ServiceNow platform, so it creates value by reducing compliance and data-breach risk rather than by direct sales. It could be monetised as a reusable ACL template or update set for educational institutions, as part of a ServiceNow implementation and consulting service, or as a managed security-configuration service sold with a subscription.
6	Scalability of the Solution	Other branches (ECE, CSE) can be supported by adding roles and data conditions, with no redesign. The same pattern works for any other table holding sensitive data. Roles can be assigned to groups instead of individual users to handle thousands of users, and the ACLs can be packaged as a scoped application or update set to move between instances. ACLs are evaluated on the server, so they work for large record volumes.